

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

Krishna Mishra (11/08/2026)



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Session	2025-26 (ODD)	Course Name	LAB:Neural Networks and Deep Learning
Semester	7th	Course Code	23ADS1704
Roll No	41	Name of Student	Krishna Mishra

Practical Number	04
Course Outcome	CO1: Understand the fundamental concepts of deep learning and neural network architectures CO2: Apply different techniques to develop and optimize deep learning models. CO3: Analyse the performance of deep learning models using appropriate evaluation techniques. CO4: Design and implement deep learning solutions for real-world problems using modern tools and frameworks.
Aim	Develop an Image Classification model using a Convolutional Neural Network (CNN).
Theory	A Convolutional Neural Network (CNN) is a deep learning model mainly used for image processing and classification. Unlike a basic neural network, CNNs automatically learn important spatial features such as edges, shapes, and textures from images. A CNN generally consists of convolutional layers, activation functions, pooling layers, and fully connected layers. The convolution layer extracts features using filters, while pooling reduces the spatial dimensions and computational complexity. The extracted features are then passed to dense layers for classification. In this practical, the Fashion-MNIST dataset containing 28×28 grayscale images is used. The CNN classifies each image into one of ten clothing categories.



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Procedure and Execution

Algorithm:

1. Import the required libraries.
2. Load the Fashion-MNIST dataset.
3. Define the ten class names.
4. Display sample training images.
5. Normalize pixel values from 0–255 to 0–1.
6. Reshape images into CNN-compatible format (28, 28, 1).
7. Create the CNN model.
8. Display the model summary.
9. Compile the model using an optimizer, loss function, and accuracy metric.
10. Train the model for 5 epochs with batch size 64.
11. Evaluate the model using test data.
12. Analyze the classification accuracy and loss.

Code:

```
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt
import numpy as np

(train_images, train_labels), (test_images, test_labels) =
datasets.fashion_mnist.load_data()

# Reshape images to include a channel dimension for CNN (28x28x1 for
grayscale)
train_images = train_images.reshape((60000, 28, 28, 1))
test_images = test_images.reshape((10000, 28, 28, 1))

# Normalize pixel values to be between 0 and 1
train_images, test_images = train_images / 255.0, test_images / 255.0

# Define class names for Fashion MNIST
class_names = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']

print(f"Shape of training images: {train_images.shape}")
print(f"Shape of training labels: {train_labels.shape}")
print(f"Shape of test images: {test_images.shape}")
print(f"Shape of test labels: {test_labels.shape}")

plt.figure(figsize=(10,10))
for i in range(25):
    plt.subplot(5,5,i+1)
```



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```
plt.xticks([])
plt.yticks([])
plt.grid(False)
plt.imshow(train_images[i], cmap=plt.cm.binary)
# The Fashion MNIST labels are single integers
plt.xlabel(class_names[train_labels[i]])
plt.show()

model = models.Sequential()
model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))

model.summary()

model.add(layers.Flatten())
model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10, activation='softmax')) # Output layer for 10 classes

model.summary()

model.compile(optimizer='adam',

loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=False),
              metrics=['accuracy'])

history = model.fit(train_images, train_labels, epochs=5,
                    validation_data=(test_images, test_labels))

plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label = 'val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.ylim([0.5, 1])
plt.legend(loc='lower right')
plt.show()

plt.plot(history.history['loss'], label='loss')
plt.plot(history.history['val_loss'], label='val_loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
```

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```
plt.legend(loc='upper right')
plt.show()
```

```
test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
print(f"\nTest accuracy: {test_acc}")
```

Output:

```
plt.xticks(())
plt.yticks(())
plt.grid(False)
plt.imshow(train_images[i], cmap=plt.cm.binary)
# The Fashion MNIST labels are single integers
plt.xlabel(class_names[train_labels[i]])
plt.show()
```





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	<pre>plt.show() plt.plot(history.history['loss'], label='loss') plt.plot(history.history['val_loss'], label='val_loss') plt.xlabel('Epoch') plt.ylabel('Loss') plt.legend(loc='upper right') plt.show()</pre> <pre>8. Evaluate the Model test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2) print(f'\nTest accuracy: {test_acc}') 313/313 - 4s - 12ms/step - accuracy: 0.9831 - loss: 0.2672 Test accuracy: 0.9831000137329182</pre>
Output Analysis	The Feed Forward Neural Network achieved a strong test accuracy of approximately 97.41% on the MNIST dataset. The visual inspection of sample predictions showed that the model correctly identified all the displayed handwritten digits.
Link of student Github profile where lab assignment has been uploaded	https://github.com/krish-na45/Neural-Network-Deep-Learning-Practicals
Conclusion	This practical successfully implemented a Feed Forward Neural Network (FFNN) for handwritten digit classification using the MNIST dataset. The process involved: Data Preparation: Loading the MNIST dataset, normalizing the pixel values of the images (scaling them from 0-255 to 0-1), and one-hot encoding the labels to prepare them for the neural network. Model Architecture: Designing an FFNN using Keras's Sequential API. This model consisted of:

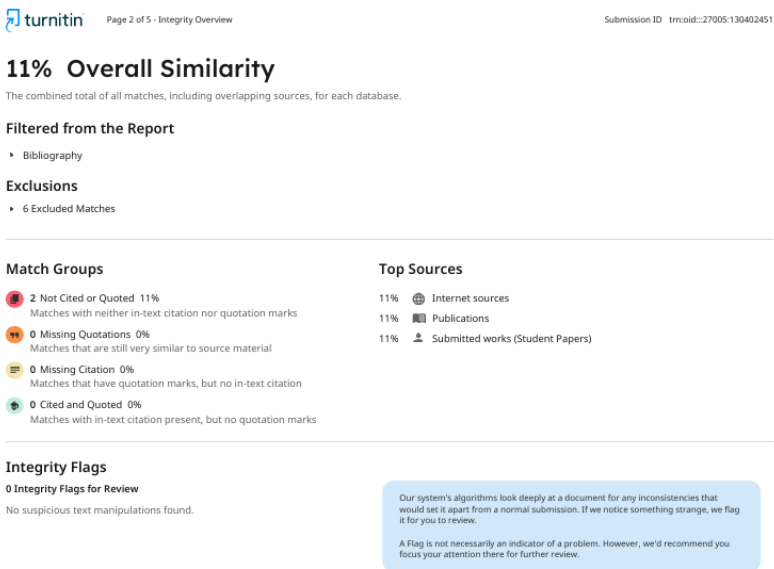
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	A Flatten layer to convert the 2D image data into a 1D vector. Two Dense hidden layers with 128 and 64 neurons respectively, both utilizing the ReLU activation function to introduce non-linearity. A Dense output layer with 10 neurons (corresponding to the 10 digit classes) and a Softmax activation function to output probability distributions. Training and Evaluation: The model was compiled with the Adam optimizer and categorical cross-entropy loss function, and then trained for 10 epochs. Post-training, the model was evaluated on an unseen test set.
Plag Report (Similarity index < 12%)	 Page 2 of 5 - Integrity Overview Submission ID: 27005:130402451 11% Overall Similarity The combined total of all matches, including overlapping sources, for each database. Filtered from the Report • Bibliography Exclusions • 6 Excluded Matches Match Groups 2 Not Cited or Quoted 11% Matches with neither in-text citation nor quotation marks 0 Missing Quotations 0% Matches that are still very similar to source material 0 Missing Citation 0% Matches that have quotation marks, but no in-text citation 0 Cited and Quoted 0% Matches with in-text citation present, but no quotation marks Top Sources 11% Internet sources 11% Publications 11% Submitted works (Student Papers) Integrity Flags 0 Integrity Flags for Review No suspicious text manipulations found. Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review. A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.
Date	11/08/2026